

Muneeb Musharaf

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EDUCATION

University College London

MSc Robotics and Artificial Intelligence

London, UK

Starting October 2026

- Selected for the **Martingale Scholarship**, a competitive, interview-based award funding the MSc in AI and Robotics at UCL.

Queen Mary University of London

BSc Computer Science and Mathematics

London, UK

Sept 2023 – July 2026

- **First Class Honours**
- **Related Modules:** Procedural & Object-oriented Programming, Automata & Formal Languages, Linear Algebra, Data structures & Algorithms, Software Engineering, Internet Protocols, Probability & Statistics, Analysis of Calculus, Numbers & Functions, Vectors & Matrices, Complex Variables, Big Data Processing, Differential Equations, Complexity & Algorithms, Bayesian Decision & Risk Analysis, Neural Networks, Complex Networks

Newham Collegiate Sixth Form Centre (A-Levels)

Mathematics (A*), Further Mathematics (A*), Physics (A)

London, UK

Aug 2021 – Aug 2023

Robert Clack School (GCSEs)

11 GCSEs, Grades 6-9

London, UK

Sept 2019 – July 2021

Maths (9), Further Maths (9), Statistics (9), Physics (8), Biology (8), Chemistry (8), English language (8), English literature (9), Computer science (9), French (6), Geography (7)

EXPERIENCE

Google DeepMind

Research Placement

London, UK

June 2025 – Aug 2025

- Selected as **1 of 12** candidates for a competitive AI research program by Google DeepMind with PhD supervision.
- Researched AI perception of music using timbre-based tasks by implementing **zero-shot and MLP experiments**.
- Resulted in co-authoring a research paper, which was accepted into **the ISMIR 2025** conference [here](#).

Data Annotation

Freelance Programmer and Annotator

London, UK

Feb 2025 – Present

- Provide expert solutions to mathematical problems and provide feedback to **train AI models** and complex data sets.
- Program projects in a variety of coding languages depending on the specifications.

Private Tutoring Practice

Self-Employed Tutor

London, UK

Aug 2023 – Present

- Deliver private one-to-one tutoring in **Mathematics, Further Mathematics, Physics and Computer Science** to GCSE and A-Level students, since finishing A-Levels in 2023.
- Provide free **personal statement guidance** for university applicants pursuing Computer Science and Mathematics-related degrees.
- Prepare students for Maths-based university entrance exams, including **TMUA, MAT and AEA**.

London Science College

Physics and Mathematics Tutor

London, UK

Sept 2023 – Jul 2026

- Delivered personalized tutoring and designed worksheets in Mathematics and Physics from GCSE to A-level.
- Ensured **90%** of students achieved high grades through tailored one-to-one and group sessions.

Queen Mary Ihsan Consulting Society

Committee Member, Marketing Manager & Tech Consulting Co-Lead

London, UK

Oct 2024

- Present

- Consult small businesses and charities on marketing strategy, managing social media campaigns through **Canva** that increased applications by **20%**.
- Co-lead the society's **tech consulting** arm alongside two peers, managing small teams to deliver software and AI solutions for client organisations.
- Led a four-person team to build an **AI-powered ethical screening tool** in **Python** for **Kestrl**, an Islamic banking app, evaluating equities against Sharia-compliant investment criteria.

My Travel World

Social Media Marketing Manager

London, UK

Aug 2024 – Aug 2025

- Implement and design visually engaging posters and digital content with **Canva** to attract more customers.
- Utilize **Google and Facebook Ads** to promote the company across social media.

Microsoft Accelerator

London, UK

Internship

Oct 2024 – Dec 2024

- Participated in Microsoft's structured mentoring programme, gaining direct exposure to industry practices in **software development and technology consulting**.
- Applied technical and collaborative skills to real-world problem-solving scenarios under professional guidance.

STEMSMART program by Cambridge University, Cavendish Labs

Cambridge, UK

Internship

Sep 2021 – May 2023

- Contributed to a two-year program on advanced topics in mathematics, physics, and business.
- Strengthened **team building** and **management skills** through collaborative projects, achieving **85%** overall.

PROJECTS

Virtual Mars Gaussian Splatting

Final Year project

Aug 2025 – May 2026

- Investigated **3D Gaussian Splatting (3DGS)** for reconstructing Martian terrain from NASA Curiosity rover imagery, benchmarking a **COLMAP** Structure-from-Motion pipeline against a CAHVOR-direct camera pose pipeline across 5 datasets.
- Diagnosed a camera intrinsic calibration error as the dominant factor limiting reconstruction quality, correcting it to raise PSNR by **~15 dB to 23.71 dB**.
- Validated pipeline generalisation across **4 new Gale Crater datasets**, producing coherent 3D reconstructions to support accessible, interactive exploration of Mars surface imagery.

Cryptocurrency Trading Platform

University Group project

Jan 2025 – Aug 2025

- Led development of a full stack platform using **Django** and **React**, implementing two-factor authentication, trading circuit breakers, portfolio tracking, and real-time API price feeds.
- Followed **Agile workflows** with **UML documentation**, focused on database integration for balances, deposits, withdrawal, and mock trade execution.

Tetris Project

Personal project

Jun 2024 – Aug 2024

- Developed a Tetris game in **Java** using the **Javax** library to deepen my understanding of Java and interface design.
- Incorporated sounds and visual effects to enhance user engagement and integrated version control with Git.

Horse Race Simulator

University project

Apr 2024

- Designed a virtual horse racing simulator in **Java** with a betting feature using virtual currency.
- Implemented a GUI with advanced features, including login verification and horse customisation for races.

ADDITIONAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, SQL, HTML/CSS, R, MATLAB
- **Frameworks & Libraries:** Django, Flask, FastAPI, React
- **AI & Data:** PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib, Neural Networks, Zero-shot Learning, Jupyter Notebooks, Google Colab
- **Tools & Platforms:** Git, GitHub, Docker, VS Code, IntelliJ, PyCharm, Visual Paradigm, PostgreSQL, MongoDB, Redis, RESTful APIs, WebSockets, Canva, Google Ads, Facebook Ads, Microsoft Office (Excel, Word, PowerPoint, Teams)
- **Operating Systems:** Windows, MacOS